

MECH 632

Intermediate Spaceflight Dynamics

Attitude Kinematics

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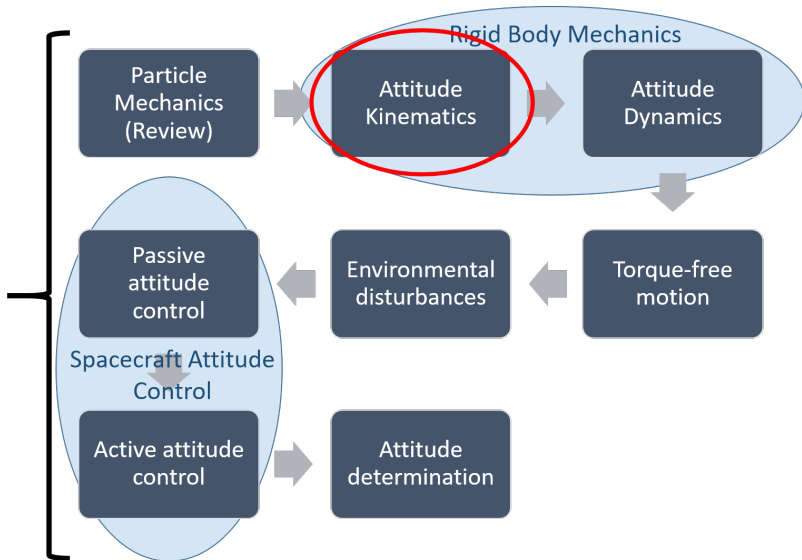
Spring 2019



Where are we?



Spacecraft Attitude Determination and Control





- 1 Lesson Goals
- 2 Attitude Parameterizations
 - Direction Cosine Matrix (DCM)
 - Euler's Axis and Angle
 - Euler Angles
 - Quaternions
 - Rodrigues Parameters (CRP and MRP)
- 3 Kinematic Differential Equations
- 4 Lesson Summary



1 Lesson Goals

2 Attitude Parameterizations

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3 Kinematic Differential Equations

4 Lesson Summary



- ▶ Understand what is the *attitude* of a rigid body
- ▶ Define the configuration space in which the attitude of a rigid body evolves, and define the properties of this configuration space
- ▶ Define Euler's Rotation Theorem
- ▶ Explore different parameterizations and understand the advantages and disadvantages of each set of parameters
- ▶ Understand what the two types of *singularities* (geometric and kinematic), and how they are exhibited in attitude parameterizations (some of them...)
- ▶ Understand the difference between *translational* kinematics and *rotational* kinematics
- ▶ Derive the general rotational kinematic equation
- ▶ Derive rotational kinematic equations for different attitude parameterizations, and be able to explain the advantages and disadvantages of each set of differential equations
- ▶ Model a spacecraft's rotational motion given angular velocity



1 Lesson Goals

2 Attitude Parameterizations

- Direction Cosine Matrix (DCM)
- Euler's Axis and Angle
- Euler Angles
- Quaternions
- Rodrigues Parameters (CRP and MRP)

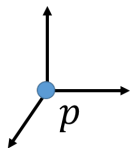
3 Kinematic Differential Equations

4 Lesson Summary

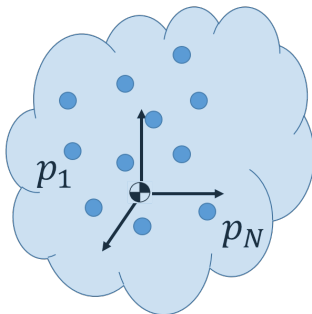
How can we model a physical system?



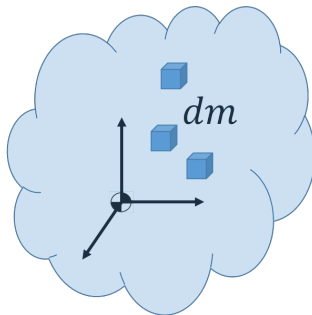
Particle



System of Particles



Rigid Body





- ▶ ...is its orientation w.r.t. some CCS
- ▶ We need to express the relationship between a body-fixed CCS and another CCS
 - ▶ What about their origin?
- ▶ Attitude $\neq$ Rotation
- ▶ Configuration space: $SO(3) = \{C \in \mathbb{R}^{3 \times 3} | C^T C = I, |C| = 1\}$
 - ▶ How many degrees-of-freedom?
- ▶ There are many ways to describe (mathematically) the attitude of a rigid body (e.g. a spacecraft)...let's explore!

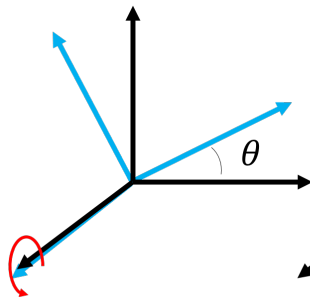


- ▶ The DCM (C_{BA}) can be used directly to define the attitude of the body-fixed CCS B relative to A ...recall $C_{BA} \in SO(3)$
 - ▶ How many elements in a DCM?
 - ▶ How many constraints? What are they?
- ▶ Advantage: C_{BA} offers an unambiguous, nonsingular, representation of attitude (i.e., there is an *isomorphism* between the physical orientation of the body and the DCM); successive rotations are easy to compute through matrix multiplication (e.g. $C_{CA} = C_{CB}C_{BA}$)
- ▶ Disadvantage: Non-linear algebraic constraints in the elements of the DCM; the elements of the DCM are not independent parameters

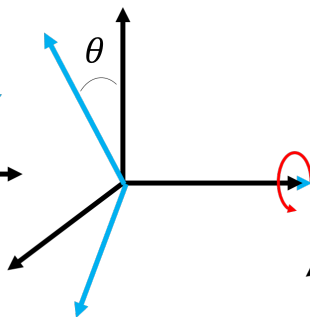
Elementary (or Principal) Rotations and their DCMs



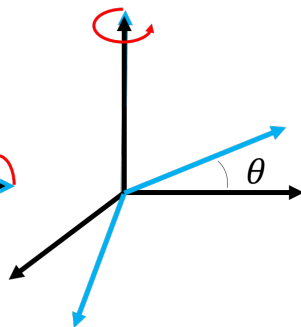
1-Rotation



2-Rotation



3-Rotation



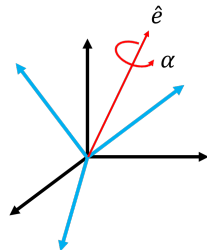
Exercise: Write the DCM for the elementary rotations, $C_1(\theta)$, $C_2(\theta)$, $C_3(\theta)$.

Euler's Axis and Angle

Euler's Eigenaxis Rotation Theorem: The general rotational displacement of a rigid body with a fixed point is a rotation about an axis ($\hat{e}$) through that point and particular angle (α).

$$\hat{e} = e_1\hat{a}_1 + e_2\hat{a}_2 + e_3\hat{a}_3$$

$$\hat{e} = e_1\hat{b}_1 + e_2\hat{b}_2 + e_3\hat{b}_3$$



- ▶ The pair $(\hat{e}, \alpha)$ can be used to describe the attitude of a rigid body
 - ▶ How many elements?
 - ▶ How many constraints? What are they?
- ▶ Advantage: nonsingular representation, only 4 parameters to keep track of
- ▶ Disadvantage: parameters are not independent, non-unique representation, $C_{BA} \implies (\hat{e}, \alpha) \implies \pm 1(\hat{e}, \alpha + 2k\pi)$ for all $k = 0, 1, 2, \dots$

Relating $(\hat{e}, \alpha)$ to the DCM

The Euler's axis (or eigenaxis): $\hat{e}^B = C_{BA}\hat{e}^A$

- ▶ What does this relationship imply?
- ▶ The DCM has 3 eigenvalues - what are they?

Extracting Euler's angle and axis from the DCM:

$$\text{tr}(C_{BA}) = C_{11} + C_{22} + C_{33} = 1 + 2 \cos \alpha$$

$$\alpha = \cos^{-1} \left(\frac{\text{tr}(C_{BA}) - 1}{2} \right) = \cos^{-1} \left(\frac{C_{11} + C_{22} + C_{33} - 1}{2} \right),$$

$$0 \leq \alpha \leq \pi \text{ (ensures uniqueness)}$$

$$\hat{e} = \frac{1}{2 \sin \alpha} \begin{bmatrix} C_{23} - C_{32} \\ C_{31} - C_{13} \\ C_{12} - C_{21} \end{bmatrix}$$

Extracting the DCM from $(\hat{e}, \alpha)$ (Euler's Formula):

$$C_{BA} = \cos \alpha I + (1 - \cos \alpha) \hat{e} \hat{e}^T - \sin \alpha [\hat{e}^\times] \quad (1)$$

Special Considerations

- ▶ $\alpha = 0 \implies C_{BA} = I \implies \hat{e}$ undefined
- ▶ $\alpha = \pi$:

$$\begin{aligned}e_1 &= \pm \sqrt{\frac{1 + C_{11}}{2}} \\e_2 &= \pm \sqrt{\frac{1 + C_{22}}{2}} \\e_3 &= \pm \sqrt{\frac{1 + C_{33}}{2}}\end{aligned}$$

with sign relationships:

$$\text{sign}(e_1) \text{sign}(e_2) = \text{sign}(C_{12})$$

$$\text{sign}(e_2) \text{sign}(e_3) = \text{sign}(C_{23})$$

$$\text{sign}(e_3) \text{sign}(e_1) = \text{sign}(C_{31})$$



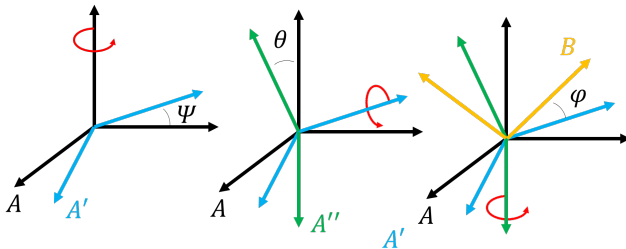
- ▶ Since the attitude of a rigid body has 3 DOF we should be able to represent it with 3 independent parameters
- ▶ Euler angles are a sequence of 3 *successive* elementary rotations to re-orient the body-fixed CCS:
 - ▶ The first rotation is about any of the three axes
 - ▶ The second rotations is about either of the two axes not used in the first rotation
 - ▶ The third rotation is about either of the two axes not used in the second rotation
- ▶ There are 12 possible Euler Angle sequences:
 - ▶ 6 symmetric: 1-2-1, 1-3-1, 2-1-2, 2-3-2, 3-1-3, 3-2-3
 - ▶ 6 asymmetric (a.k.a. Tait-Bryan angles): 1-2-3, 1-3-2, 2-1-3, 2-3-1, 3-1-2, 3-2-1
- ▶ For a particular attitude each sequence will give a *unique* set of Euler Angles
- ▶ Once a sequence is chosen it must be used consistently

The 3-2-1 Sequence

- ▶ First rotate about the 3-axis through ψ (yaw)
- ▶ Then rotate about the “new” 2-axis through θ (pitch)
- ▶ Finally rotate about the “new” 1-axis through ϕ (roll)

$$C_{BA} = C_1(\phi)C_2(\theta)C_3(\psi) \quad (2)$$

Note: The order is important!





Consider the 3-2-1 Sequence above with $\theta = \pi/2$, resulting in:

$$C_{BA} = \begin{bmatrix} 0 & 0 & -1 \\ \sin(\phi - \psi) & \cos(\phi - \psi) & 0 \\ \cos(\phi - \psi) & -\sin(\phi - \psi) & 0 \end{bmatrix}$$

The angles ϕ and ψ are indistinguishable!

- ▶ All Euler Angle sequences exhibit this behavior; all in the second rotation angle
- ▶ For asymmetric sequences: when the second angle is $\pi/2$
- ▶ For symmetric sequences: when the second angle is 0
- ▶ Singularities are the main disadvantage of using Euler Angles
- ▶ Singularities could be avoided in some situations (e.g., when only small deviations from a particular orientation are of interest)



- ▶ We have seen how to compute the DCM corresponding to a particular Euler angle sequence (e.g., 3-2-1)
- ▶ Extracting the Euler Angles from a DCM is again sequence dependent
- ▶ The 3-2-1 Euler angles can be extracted by:

$$\phi = \tan^{-1}(C_{23}/C_{33})$$

$$\theta = \sin^{-1}(-C_{13})$$

$$\psi = \tan^{-1}(C_{12}/C_{11})$$

Note: Make sure you use the `atan2` command in MATLAB to avoid quadrant ambiguity



Advantages:

- ▶ Minimal parametrization (only 3 independent parameters)
- ▶ Intuitive to visualize

Disadvantages:

- ▶ Singularities! Large angle maneuvers may be difficult to represent
- ▶ Trig functions in the parameters may increase computational cost (transcendental expressions)



- ▶ Recall Euler's rotation theorem...any rigid body rotation can be represented by $(\hat{e}, \alpha)$
- ▶ The unit *quaternion* is a set of 4 parameters ($q = [q_1, q_2, q_3, q_4]^T$), made up of a vector part ($\bar{q} = [q_1, q_2, q_3]^T$) and a scalar part (q_4) defined as:

$$\begin{aligned}\bar{q} &= q_{1:3} = \hat{e} \sin(\alpha/2) \\ q_4 &= \cos(\alpha/2)\end{aligned}$$

- ▶ As a 4 parameter set the quaternion is constrained by the following:

$$\|q\| = q_1^2 + q_2^2 + q_3^2 + q_4^2 = \bar{q}^T \bar{q} + q_4^2 = 1$$

Note: This constraint is a 4-D unit sphere; any rotation described by a quaternion has a trajectory on the *surface* of this sphere

Leonhard Euler (1707-1783):



- ▶ Did just about everything...
- ▶ Arguably his most remarkable contribution is Euler's formula, which provided a simple way to multiply complex numbers:

$$e^{i\theta} = \cos \theta + i \sin \theta$$

Sir William Rowan Hamilton (1805-1865):



- ▶ Inventor of the quaternion; in search of hypercomplex numbers

$$i^2 = j^2 = k^2 = ijk = -1$$

- ▶ Pure mathematician; although his work plays a major role in physics & engineering

Algebraic operations with quaternions:

For $q, w \in \mathbb{H}$,

Addition: $q + w = (\bar{q} + \bar{w}, q_4 + w_4)$

Scaling: $\alpha q = (\alpha \bar{q}, \alpha q_4)$

Multiplication: $q \otimes w = (q_4 \bar{w} + w_4 \bar{q} + \bar{q} \times \bar{w}, q_4 w_4 - \bar{q}^T \bar{w})$

Conjugation: $q^* = (-\bar{q}, q_4)$

Norm: $\|q\|^2 = q^* \otimes q = (\bar{0}, q_4^2 + \bar{q}^T \bar{q})$



- Ambiguity: Given a certain attitude there are two sets of quaternions that describe the same orientation

$$q = q(\hat{e}, \alpha) \iff -q = q(-\hat{e}, -\alpha)$$

- These are antipodal points on the 4-D unit sphere; one of these points specifies the *shortest* single axis rotation and the other is the *longest*
- The shortest rotation ($\alpha \leq \pi$) corresponds to the quaternion with positive scalar part ($q_4 > 0$)
- Some references use different conventions (i.e. Wie uses $q = [q_1, q_2, q_3, q_4]^T$, while Schaub/Junkins use $\beta = [\beta_0, \beta_1, \beta_2, \beta_3]^T$)
 - We will stick with Wie's convention (fairly common)
 - MATLAB's Aerospace toolbox uses the other convention
 - Be careful when/if comparing results



- ▶ The DCM in terms of the quaternion can be derived by Euler's formula (Slide 12):

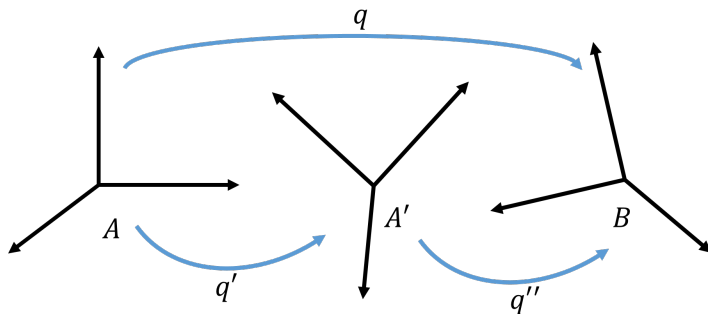
$$C_{BA} = (q_4^2 - \bar{q}^T \bar{q})I + 2\bar{q}\bar{q}^T - 2q_4[\bar{q}^\times] \quad (3)$$

Note that q and $-q$ produce the same DCM...

- ▶ The quaternion can be extracted from the DCM by:

$$q_4 = \pm \frac{1}{2} \sqrt{1 + C_{11} + C_{22} + C_{33}}$$
$$\bar{q} = \frac{1}{4q_4} \begin{bmatrix} C_{23} - C_{32} \\ C_{31} - C_{13} \\ C_{12} - C_{21} \end{bmatrix}$$

- ▶ The positive sign is kept for q_4 to avoid the ambiguity issue (it is important to stay consistent)
- ▶ Issue when $q_4 \rightarrow 0$: see Sheppard's algorithm in Schaub and Junkins, Section 3.5



Using the DCM successive rotations:

$$q' \rightarrow C_{A'A}, \quad q'' \rightarrow C_{BA'}, \quad q \rightarrow C_{BA} = C_{BA'}C_{A'A}$$
$$q = q'' \otimes q'$$

Quaternion multiplication is a *bilinear* transformation; associative but not commutative

$$q = q'' \otimes q' = \begin{bmatrix} q_4'' & q_3'' & -q_2'' & q_1'' \\ -q_3'' & q_4'' & q_1'' & q_2'' \\ q_2'' & -q_1'' & q_4'' & q_3'' \\ -q_1'' & -q_2'' & -q_3'' & q_4'' \end{bmatrix} q' = G(q'')q'$$

or equivalently,

$$q = \begin{bmatrix} q_4' & -q_3' & q_2' & q_1' \\ q_3' & q_4' & -q_1' & q_2' \\ -q_2' & q_1' & q_4' & q_3' \\ -q_1' & -q_2' & -q_3' & q_4' \end{bmatrix} q'' = G(q')^T q''$$

Note: $G(q)$ is an orthonormal matrix and $GG^T = I$



Advantages:

- ▶ Quaternions are the most widely used attitude parametrization method
- ▶ No singularities
- ▶ Quaternion multiplication can be expressed as matrix multiplication with no trig functions

Disadvantages:

- ▶ Unit norm constraint must be enforced
- ▶ Difficult to visualize

Classical Rodrigues Parameters:

$$\mathbf{g} = \frac{\bar{q}}{q_4} = \hat{e} \tan \alpha/2$$

- ▶ Singularity at $\alpha = \pm\pi$
- ▶ Minimal representation (3 parameters)
- ▶ $\mathbf{g}(q) = \mathbf{g}(-q)$; same ambiguity as seen in the quaternion parameterization

Modified Rodrigues Parameters:

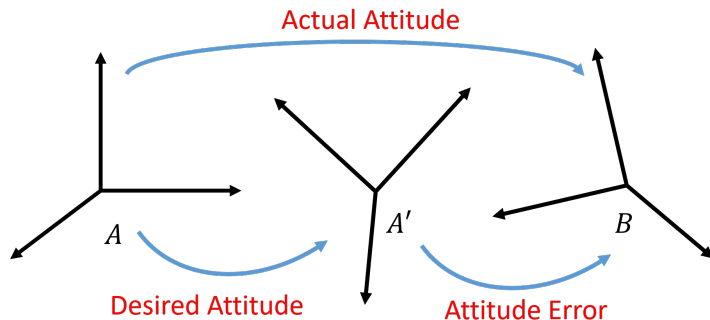
$$\boldsymbol{\sigma} = \frac{\bar{q}}{1 + q_4} = \hat{e} \tan \alpha/4$$

- ▶ Singularity at $\alpha = \pm 2\pi$
- ▶ Minimal representation (3 parameters)
- ▶ $\boldsymbol{\sigma}(q) \neq \boldsymbol{\sigma}(-q)$; the *same* attitude can be represent by two *different* sets of MRPs
- ▶ Referred to as the shadow MRP, $\boldsymbol{\sigma}^s$



- ▶ Most commonly used attitude parameterizations:
 - ▶ Euler Angles (intuitive, useful for motion around “stable” orientations”)
 - ▶ Quaternions (nonsingular and low computational complexity, useful for large rotations)
 - ▶ We will mostly use these two parameterizations in this class
- ▶ CRPs and MRPs are gaining more popularity for some applications
- ▶ Other attitude parameterizations we will not cover: stereographic parameters, higher order RPs, Cayley-Klein parameters...

Attitude Error: the difference between a desired attitude and the actual attitude; in the configuration space $SO(3)$ it is not a simple vector subtraction (i.e., $C_{\text{error}} \neq C_{\text{desired}} - C_{\text{actual}}$)



$$C_e = C_{BA}C_{A'A}^T \implies q_e = q \otimes q_d^*$$

Note: $q^* = [-\bar{q}, q_4]^T$



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- ▶ In a Euclidian space ($\mathbb{R}^3$) the following translational kinematic relationship holds:

$$\dot{\vec{r}} = \vec{v}$$

- ▶ In $SO(3)$ however:

$$\dot{C} \neq \vec{\omega}$$

- ▶ We need to derive the appropriate *kinematic differential equations* in order to describe the rotational motion of the body-fixed CCS w.r.t. to another CCS

The DCM Kinematic Equation



The instantaneous angular velocity of B with respect to A is:

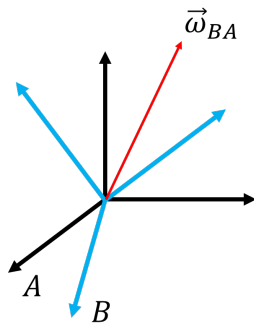
$$\vec{\omega}_{BA} = \omega_1 \hat{b}_1 + \omega_2 \hat{b}_2 + \omega_3 \hat{b}_3$$

By the definition of the DCM:

$$\begin{bmatrix} \hat{b}_1 \\ \hat{b}_2 \\ \hat{b}_3 \end{bmatrix} = C_{BA} \begin{bmatrix} \hat{a}_1 \\ \hat{a}_2 \\ \hat{a}_3 \end{bmatrix} \Rightarrow \begin{bmatrix} \hat{a}_1 \\ \hat{a}_2 \\ \hat{a}_3 \end{bmatrix} = C_{BA}^T \begin{bmatrix} \hat{b}_1 \\ \hat{b}_2 \\ \hat{b}_3 \end{bmatrix}$$

Taking the time derivative w.r.t A results in the kinematic equation:

$$\dot{C}_{BA} = -[\vec{\omega}_{BA}^\times] C_{BA} \quad (4)$$





By rearranging Eq. 4 we can derive the following relationships:

$$\begin{aligned}\omega_1 &= \dot{C}_{21}C_{31} + \dot{C}_{22}C_{32} + \dot{C}_{23}C_{33} \\ \omega_2 &= \dot{C}_{31}C_{11} + \dot{C}_{32}C_{12} + \dot{C}_{33}C_{13} \\ \omega_3 &= \dot{C}_{11}C_{21} + \dot{C}_{12}C_{22} + \dot{C}_{13}C_{23}\end{aligned}\tag{5}$$

These expressions will be useful in deriving kinematic relationships for other attitude parameters.

Note: These are the scalar components of $\vec{\omega}_{BA}$ in the B CCS.

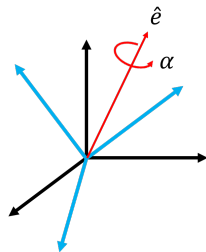
The $(\hat{e}, \alpha)$ Kinematic Equations

$$\begin{aligned}\vec{\omega}_{BA} &= \dot{\alpha} \hat{e} \implies \\ \dot{\alpha} &= \hat{e}^T \vec{\omega}_{BA}\end{aligned}$$

Using the relationships in Eq. 5 with the DCM relationship in Eq. 1,

$$\dot{\hat{e}} = \frac{1}{2} \left[[\hat{e}^\times] - \cot\left(\frac{\alpha}{2}\right) [\hat{e}^\times]^2 \right] \omega_{BA}^B$$

- ▶ Kinematic singularity: what happens at $\alpha = 0$?
- ▶ These equations are rarely used in practice...

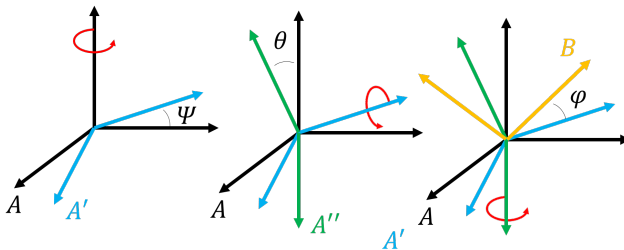


Euler Angle Kinematic Equations (3-2-1)



Recall:

$$C_{BA} = C_1(\phi)C_2(\theta)C_3(\psi)$$



$$\begin{aligned}\vec{\omega}_{A'A} &= \dot{\psi}\hat{a}_3 = \dot{\psi}\hat{a}'_3 \\ \vec{\omega}_{A''A'} &= \dot{\theta}\hat{a}''_2 = \dot{\theta}\hat{a}'_2 \\ \vec{\omega}_{BA''} &= \dot{\phi}\hat{b}_1 = \dot{\phi}\hat{a}''_1\end{aligned}$$



The total angular velocity is:

$$\begin{aligned}\vec{\omega}_{BA} &= \vec{\omega}_{BA''} + \vec{\omega}_{A''A'} + \vec{\omega}_{A'A} \\ &= \dot{\phi}\hat{b}_1 + \dot{\theta}\hat{a}_2'' + \dot{\psi}\hat{a}_3'\end{aligned}$$

Let's express all quantities in the B CCS:

$$\begin{aligned}\omega_{BA}^B &= \begin{bmatrix} \dot{\phi} \\ 0 \\ 0 \end{bmatrix} + C_1(\phi) \begin{bmatrix} 0 \\ \dot{\theta} \\ 0 \end{bmatrix} + C_1(\phi)C_2(\theta) \begin{bmatrix} 0 \\ 0 \\ \dot{\psi} \end{bmatrix} \\ \omega_{BA}^B &= \begin{bmatrix} 1 & 0 & -\sin \theta \\ 0 & \cos \phi & \sin \phi \cos \theta \\ 0 & -\sin \phi & \cos \phi \cos \theta \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = S \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}\end{aligned}$$



Rearranging terms:

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = S^{-1} \omega_{BA}^B \quad (6)$$

where

$$S^{-1} = \frac{1}{\cos \theta} \begin{bmatrix} \cos \theta & \sin \phi \sin \theta & \cos \phi \sin \theta \\ 0 & \cos \phi \cos \theta & -\sin \phi \cos \theta \\ 0 & \sin \phi & \cos \phi \end{bmatrix}$$

- ▶ We already saw that Euler angles exhibit a geometric singularity
- ▶ Kinematic singularity: what happens at $\theta = \pi/2$?
- ▶ These singularities could cause issues when numerically integrating the Euler angles

Using the angular velocity/DCM relationships derived in Eq. 5, and the quaternion parameterization of the DCM in Eq. 3:

$$\begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} = \begin{bmatrix} \left(\frac{\partial C_{21}}{\partial q}\right)^T C_{31} + \left(\frac{\partial C_{22}}{\partial q}\right)^T C_{32} + \left(\frac{\partial C_{23}}{\partial q}\right)^T C_{33} \\ \left(\frac{\partial C_{31}}{\partial q}\right)^T C_{11} + \left(\frac{\partial C_{32}}{\partial q}\right)^T C_{12} + \left(\frac{\partial C_{33}}{\partial q}\right)^T C_{13} \\ \left(\frac{\partial C_{11}}{\partial q}\right)^T C_{21} + \left(\frac{\partial C_{12}}{\partial q}\right)^T C_{22} + \left(\frac{\partial C_{13}}{\partial q}\right)^T C_{23} \end{bmatrix} \dot{q} \quad (7)$$

with $q = [q_1 \ q_2 \ q_3 \ q_4]^T$ yields,

$$\begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} = 2 \begin{bmatrix} q_4 & q_3 & -q_2 & -q_1 \\ -q_3 & q_4 & q_1 & -q_2 \\ q_2 & -q_1 & q_4 & -q_3 \end{bmatrix} \dot{q}$$

But...how can we invert this expression?

Recall the constraint equation:

$$\begin{aligned} q_1^2 + q_2^2 + q_3^2 + q_4^2 &= 1 \implies \\ 2(q_1\dot{q}_1 + q_2\dot{q}_2 + q_3\dot{q}_3 + q_4\dot{q}_4) &= 0 \end{aligned}$$

Appending this equality to the kinematic equation:

$$\begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \\ 0 \end{bmatrix} = 2 \begin{bmatrix} q_4 & q_3 & -q_2 & -q_1 \\ -q_3 & q_4 & q_1 & -q_2 \\ q_2 & -q_1 & q_4 & -q_3 \\ q_1 & q_2 & q_3 & q_4 \end{bmatrix} \dot{q}$$

Note that this 4×4 matrix is orthonormal!

$$\dot{q} = \frac{1}{2} \begin{bmatrix} q_4 & -q_3 & q_2 & q_1 \\ q_3 & q_4 & -q_1 & q_2 \\ -q_2 & q_1 & q_4 & q_3 \\ -q_1 & -q_2 & -q_3 & q_4 \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \\ 0 \end{bmatrix}$$

or more compactly,

$$\dot{q} = \frac{1}{2} \begin{bmatrix} [\bar{q}^\times] + q_4 I \\ -\bar{q}^T \end{bmatrix} \omega_{BA}^B \quad (8)$$

- ▶ This differential equation contains no singularities and no trig functions (computationally efficient)
- ▶ Most widely used method to numerically integrate rigid body attitude



- ▶ The process we used to derive the quaternion kinematic equation can be used with any set of parameters (Eq. 7)
- ▶ Similar expressions can be derived for CRP and MRP representations (see Bong Wie or Schaub/Junkins if interested)
- ▶ We will focus on Euler angle and quaternion representations (mostly) in this class



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- ▶ We saw a few different sets of parameters which we can represent the attitude of a spacecraft
 - ▶ *Redundant* sets have more parameters than needed and parameters have dependencies (constraints)
 - ▶ *Minimal* sets have independent parameters but exhibit geometric singularities
- ▶ We derived kinematic equations for the rotational motion of a rigid body using different parameterizations
- ▶ Given an angular velocity profile we can now model the attitude motion of a spacecraft!